Machine Learning Approaches for Arabic NLP

Authors: Dr. Tariq Al-Mansoor, Prof. Elena Rostova

Abstract

This paper investigates transformer architectures for morphological disambiguation.
xd9x8axd8xaaxd9x86xd8xa7xd9x88xd9x84 xd9x87xd8xb0xd8xa7 xd8xa7xd9x84xd8xa8xd8xadxd8xab xd9x86xd9x85xd8

1. Methodology and Model Architecture

We utilized bidirectional cross-attention with 24 layers and 16 attention heads.

Training was conducted on a multi-genre Arabic dataset comprising 50M tokens.

Table 1: Hyperparameter Configuration

Parameter	Layer Count	Learning Rate	Batch Size
Value	24	1e-4	128

2. Results and Conclusion

The proposed model achieved 94.8% F1-score across benchmark datasets.

xd8xa3xd8xb8xd9x87xd8xb1xd8xaa xd8xa7xd9x84xd9x86xd8xaaxd8xa7xd8xa6xd8xac xd8xaaxd9x81xd9x88xd9x82xd8x

References

[1] Vaswani et al., Attention Is All You Need, NeurIPS 2017.

[2] Al-Badr et al., Arabic Semantic Benchmarks, ACL 2024.